Agentic AI Driven Network Management Migration Challenges and Opportunities in IETF

draft-wmz-nmrg-agent-ndt-arch-03

IETF 125 Meeting 03/17/2026

on behalf of Authors Team (Qin Wu,
Luis M. Contreras, Cheng Zhou, Sai Han, Yong-Geun Hong,Qiufang Ma,
Zhengqiang Li, Lionel Taihardat)

Agentic AI Evolution: Moving from Automation to Autonomy

AN Level Evolution pace

Level 3 (before 2024)
Machine-assisted network automation

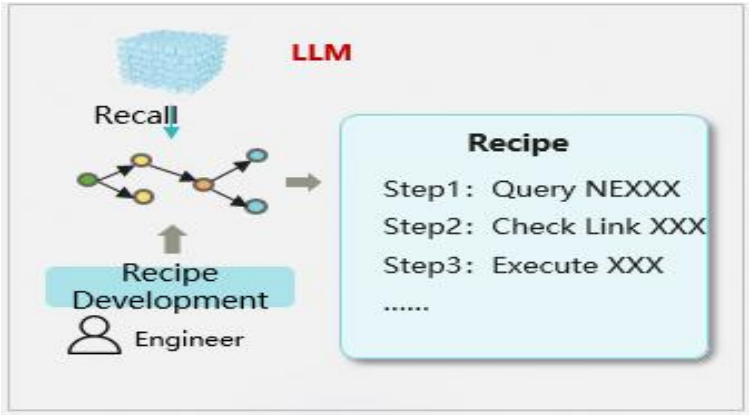
Phrase 1 of Level 4 (2025 to 2027)
Closed loop of single-domain network autonomy enabled by agents

Phrase 2 of Level 4 (2028 to 2030)
Multiple agents collaborate with each other for E2E closed loop autonomy.

We are here

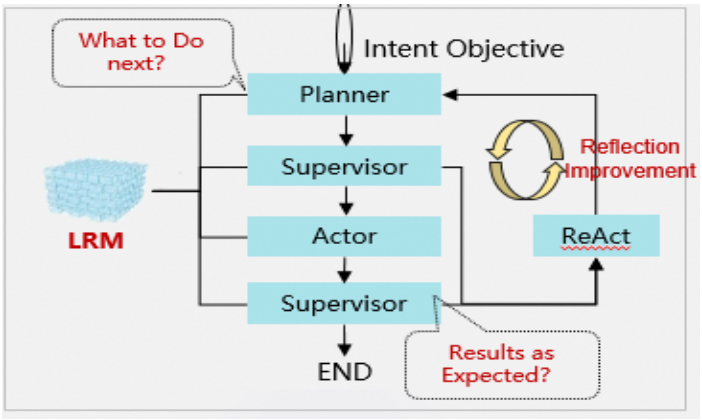
Agentic AI Evolution pace

Agent 1.0



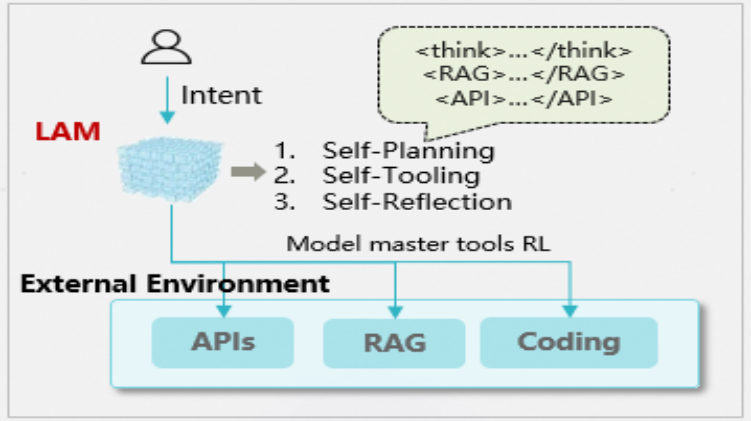
1.Support Recipe/API retrieval, & parameter filling
2.Supports document parsing and Q&A.

Agent 2.0



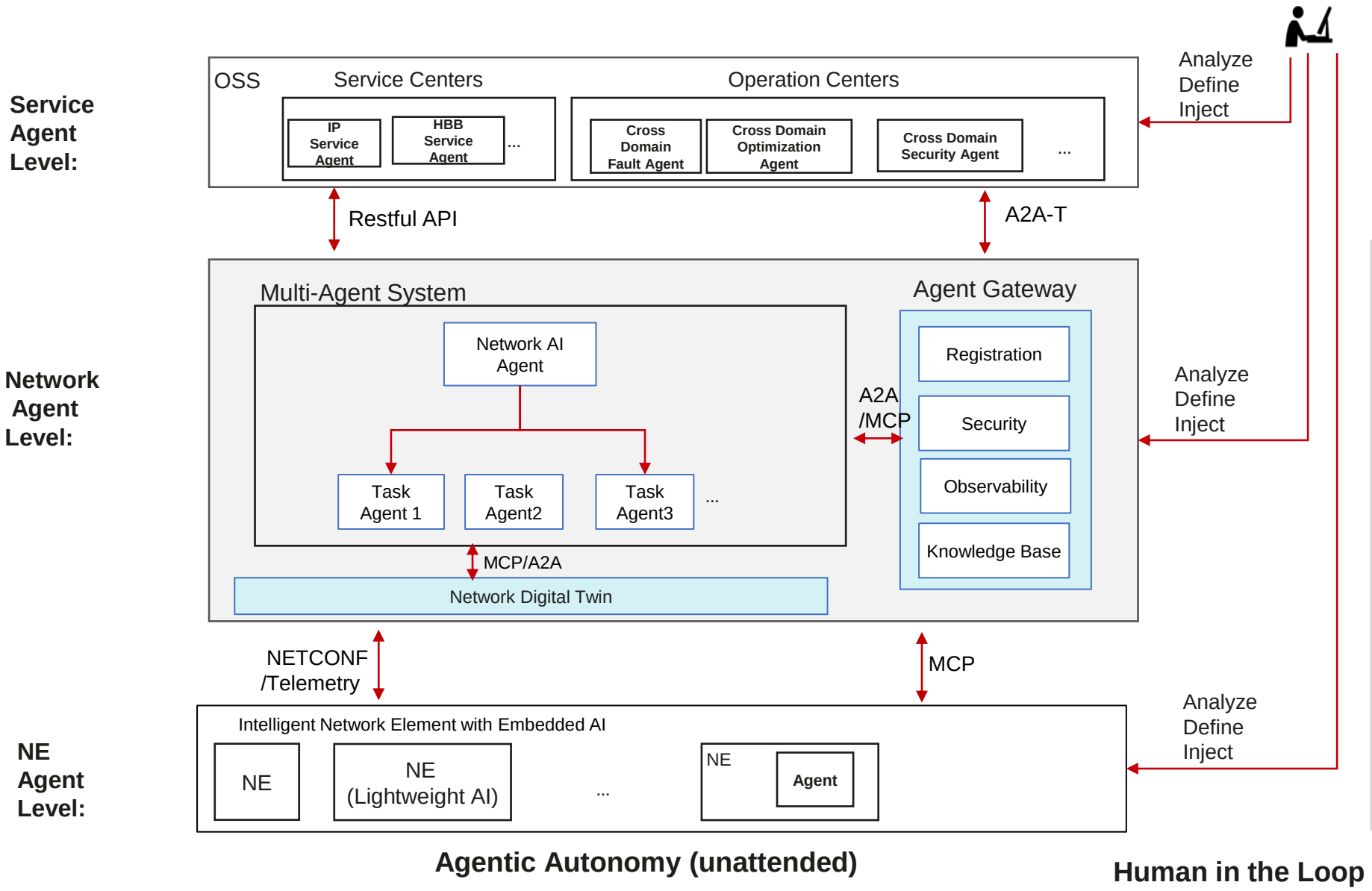
1. Support Planning+ReAct
2. Support functioncall+MCP， CodeAct

Agent 3.0



1.Self-planning/tooling/reflection
2.Train Agentic Model with Agentic Capabilities

Agentic AI IP Network Management Architecture



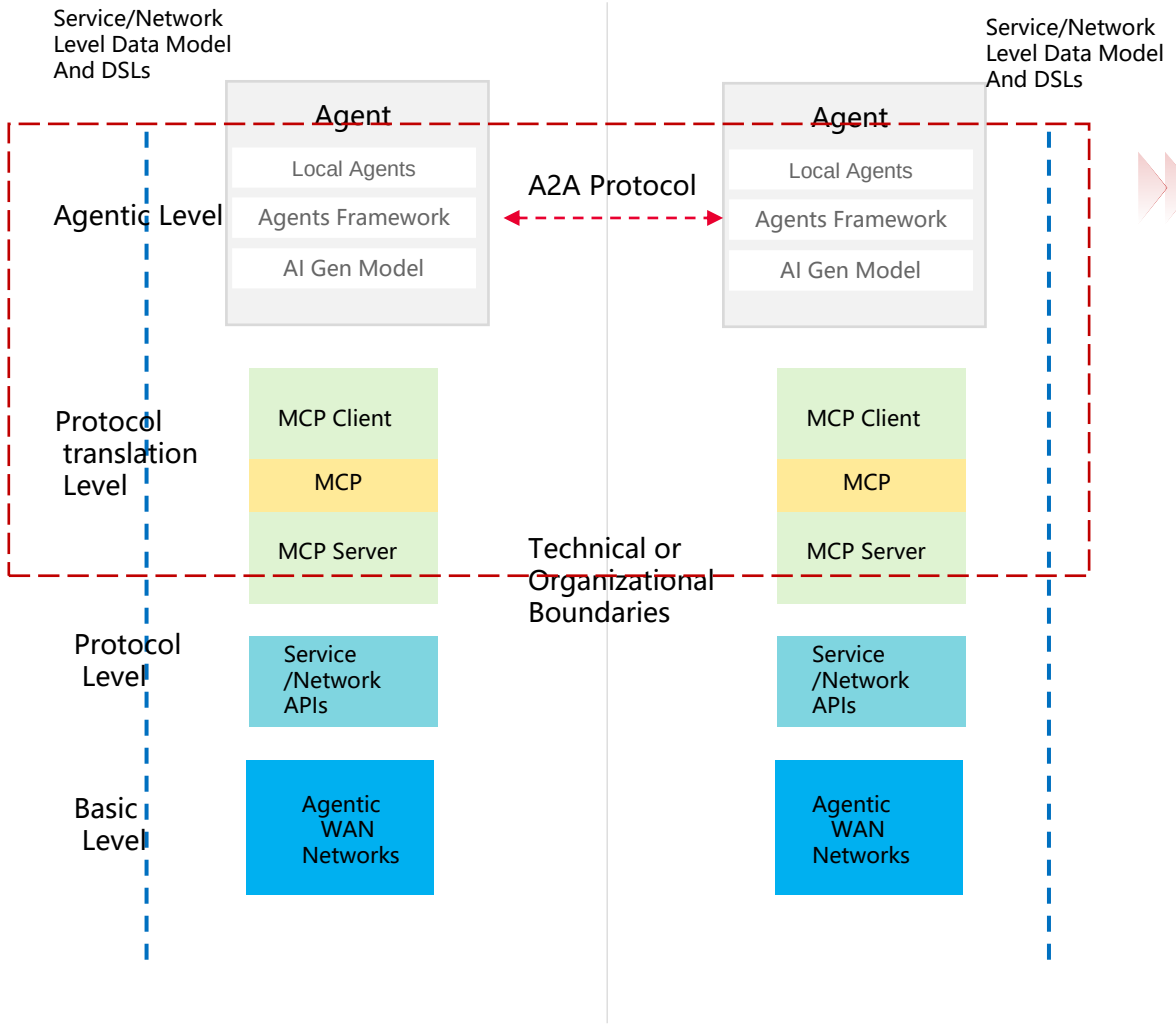
Architecture Evolution

- 3 Level Agentic Architecture
- Single Agent, Single Scenario=> Multi-Agent, Multi-Scenario Collaboration In the Autonomous Domain

Protocol Evolution

- Service AI Agent and Network AI Agent Interaction
  - Restulf API=>A2A-T
- Network AI Agent and NE AI Agent Interaction
  - NETCONF/Telemetry Coexist with MCP
- Human and Agent interaction through Agent Gateway
  - Human in the loop interface

Key Challenges for Agentic AI for IP network Management



Challenge1: DSL for Agents and Models

- **Lack Semantic & Contextual info** for AI to reason about Intent
  - Natural language interactions need to handle uncertain or ambiguous tasks
- **Data Quality and Data Silo**
  - Difficult to generalize across different device types or data sources with
    - High frequency collected data
    - Standardized format
- **Significant shortage of professionals** with the cross-disciplinary expertise
  - For both Network Development and AI development

Legend:
OBO: On behalf Of
DSL: Domain Specific Language

Domain Specific Language for Agents and models

Solution to Challenge 1:

Service/Network Level Payload

L3VPN Service Model
L2VPN Service Model
Network Slicing Service Model
Attachment Circuit Service Model
TE Service Mapping Model

L3VPN Network Model
L2VPN Network Model
Network Slicing Resource Model
Attachment Circuit Network Model
Service Attachment Point Network Model

Network Topology Base Model
L3 Topology Network Model
L2 Topology Network Model
TE Topology Network Model
DC Fabric Topology Network Model
Network Slice Topology Network Model

DSL for Agents and Models

Ontology- Entities (e.g.,L3VPN Connectivity, L2VPN Connectivity, SAP,ACaaS,etc)

Semantics- Properties of Entities (e.g., access-type, site-location)

Constraints- Value range, enumerations and relationship

Context- How one object is related to others (e.g., SLE associated with NS connectivity)

DSL Payload, APIs

Service/Network Specific YAML (Payload)

IETF Network/Service Specific APIs

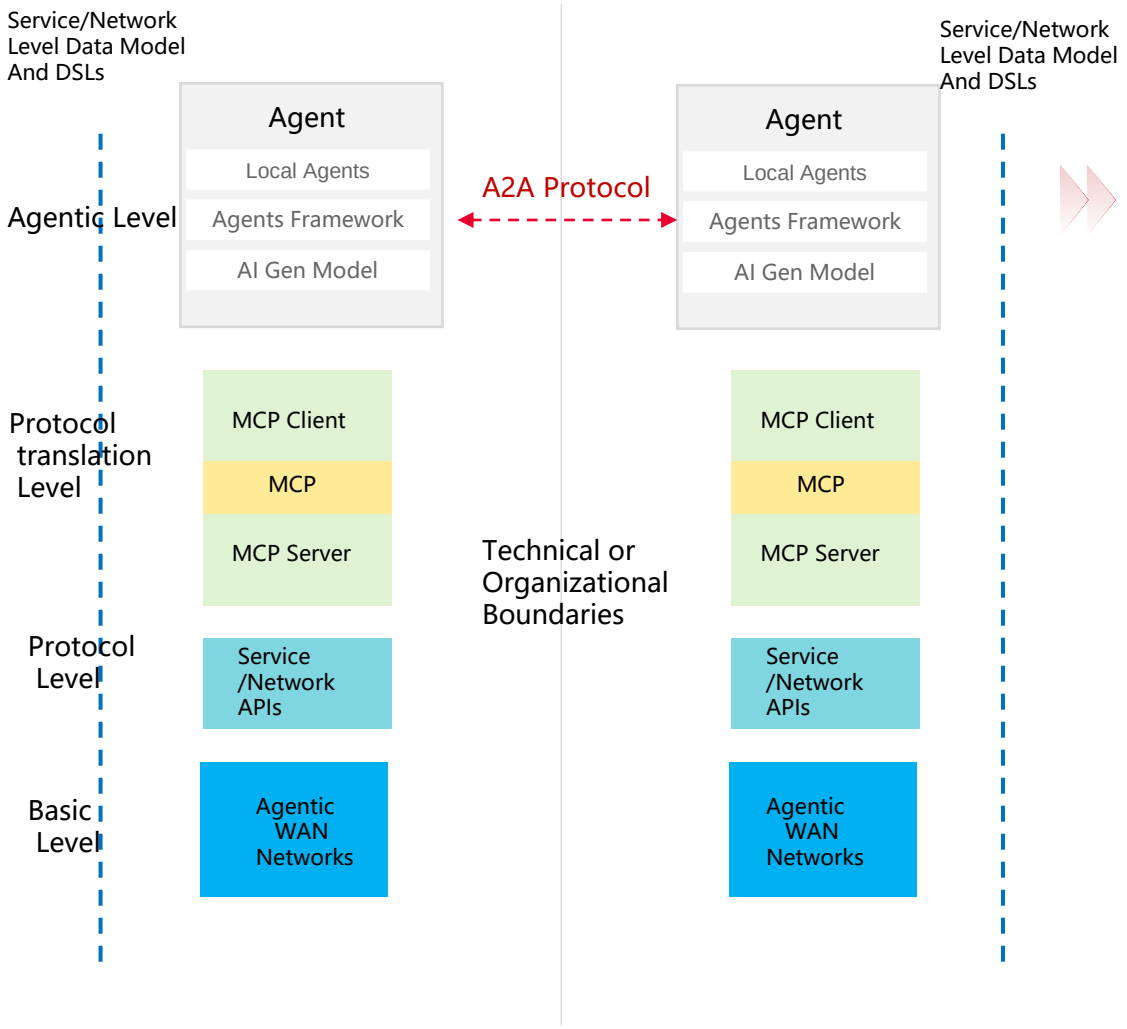
DSL SDK

MCP Server

Candidate IETF work:

- DSL payload documentation For all Service/Network Level Model
- IETF YANG generated API Profile documentation

Key Challenges for Agentic AI for IP network Management



Challenge2 : Protocol among Agents/Tools/Models

- High Risk Operations
  - E.g., "bulk deletion" or "core route changes“
- The Collaboration Timeliness Requirements
  - Lack support for a native pub/sub or event-driven
  - Slow reason speed vs real time management operation such as fault management
- The Collaboration Reliability
  - Complex task with high level expertise
  - Reliable input with cleaned, validated data

Legend:
OBO: On behalf Of

Separate Protcol from Prompt Language Template for Agents and models

Solution to Challenge 2:

A2A-T Protocol Family

IETF,etc 3rd protocol Ext

Optional Capability: A2A-T Negotiation | A2A-T Authorization

Mandatory Capability: A2A-T Task Ext | A2A-T Notification

Agent Discovery(ID) | Ext Capability Discovery | Skill Discovery

A2A Base Protocol

Protocol ToolChain

A2A-T toolchain

One Click Deployment A2A-T SDK | Representative case Example Code

A2A Tool Chain (Open Source Protocol、AgentSDK、example Code,etc)

LF: Ext Capability Register

Linux A2A-T

Cross SDO Coordination

TMF | IETF | 3GPP

APTL Model Family

DSL Model: DSL Templ (FM) | DSL Tem (Opti | DSL Temp (qualty) | DSL Templa () | DSL Template ()

Base Capability: DSL Template Capability Discovery

APTL MetaModel Defintion（Base Standard）

Model ToolChain

IETF YANG Interface to Template Tool

TMF intent Interface to Template Tool

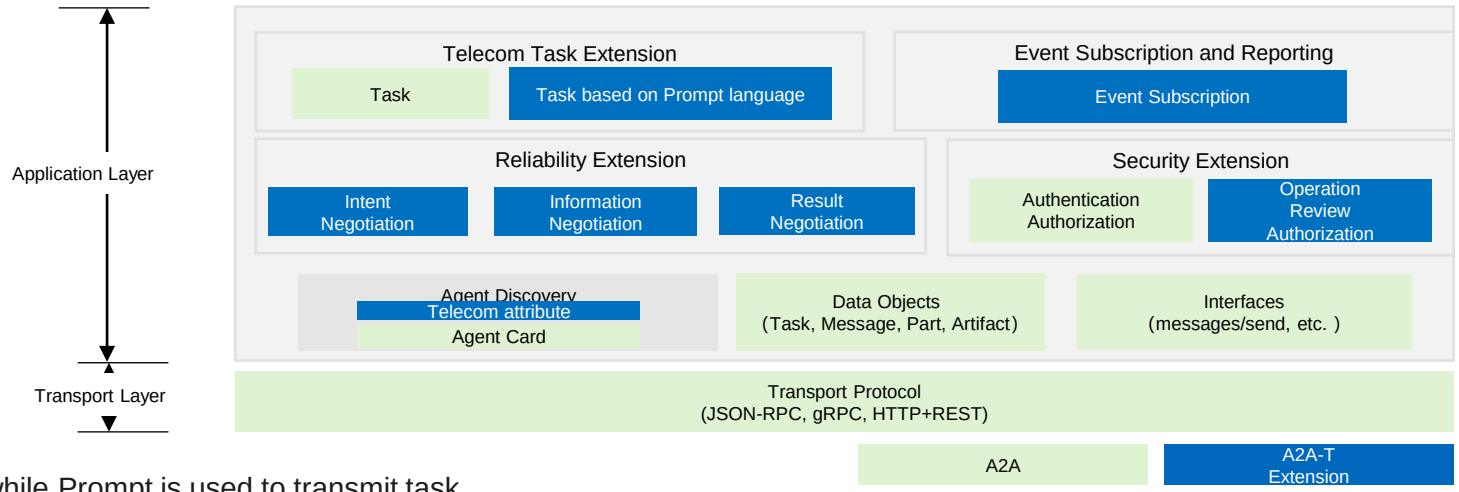
3GPP Intent Interface to Template Tool

Candidate IETF work:

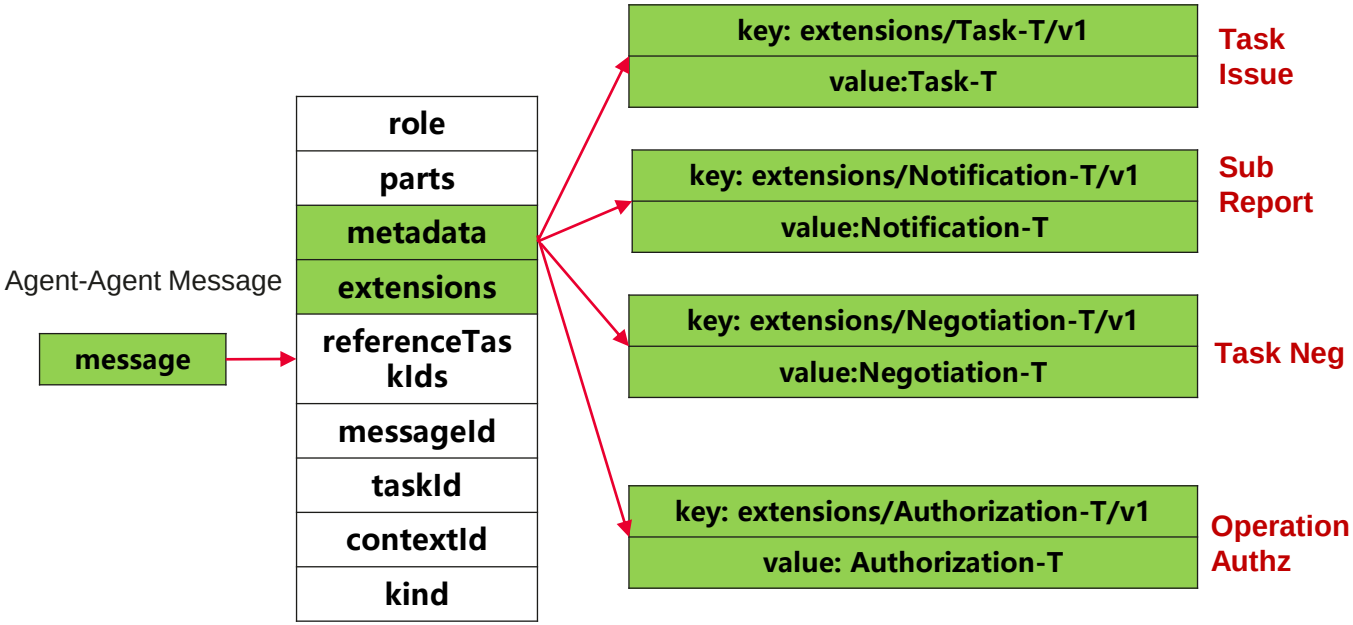
- Define semantic information transfer or agent prompt language template for IETF intent
  - Explore how IETF intent can be carried as YANG Payload over A2A-T
  - Ensure the accuracy and efficiency of structured data for deterministic tasks

A2A Extension A2A-T Overview：TMF structured Prompt Templates

Solution to Challenge 3:



Agent-Agent collaboration is based on the A2A-T, while Prompt is used to transmit task instructions and info, including task issuance, network maintenance subscription reporting, task negotiation, and operation authorization



Task-T：Prompt Template

L0 Layer Task Prompt Template (1 template)
L1 Layer Task Prompt Templates (6 templates): Fault Diagnosis Task, Fault Repair Task, Complaint Diagnosis Task, Task Result Report Download, Energy Efficiency Optimization Task, Network Quality Optimization Task

Notification-T：Prompt Template

Layer 0 Event Subscription and Reporting Template (1 template)
Layer 1 Event Subscription and Reporting Template (4 templates): Incident Event Subscription and Reporting, Energy Efficiency Optimization Event Subscription and Reporting, Network Quality Optimization Event Subscription and Reporting, Home Broadband Experience Event Subscription and Reporting

Negotiation-T：Prompt Template

4 templates: Clarification of Intent, Clarification of Feasibility, Negotiation of Outcomes, and Negotiation of Information.

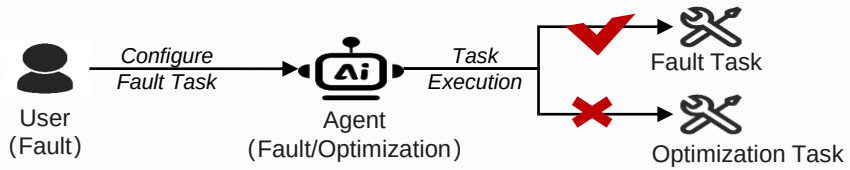
Authorization-T：Prompt Template

3 templates: Configure full operation authorization rules, Modify partial operation authorization rules, and Request operation authorization during task execution.

A2A-T Prompt Templates for Task, Notification, Negotiation, Authorization can be reused

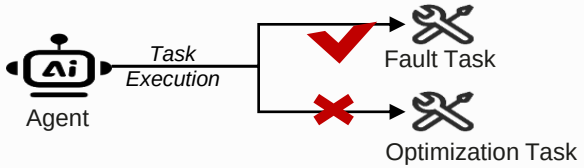
Key Challenges for Agentic AI for IP network Management

Users access through a multi-functional agent



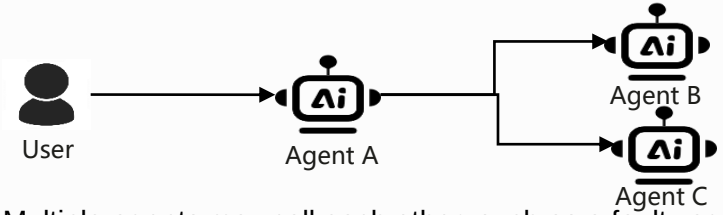
- Restrict users from using non-user functions through the multi-functional intelligent agent (fault and optimization)

System's self-manage agent automatically triggered tasks



- In single-agent scenario, the agent triggers tasks automatically based on system monitoring and other metrics, restricting access scope.

Cascading access of agents in a multi-agent architecture



- Multiple agents may call each other, such as a faulty agent A calling an optimizing agent B, which requires access control.

Challenge3: Trust and Security for Agents

- OBO Human
  - Human Manipulation
  - Agent Delegation or authorization
- OBO Tools/APIs/LLMs
  - Misuse of Tools
  - Memory Poison
  - Resource overloading
- OBO other AI Agents
  - Repudiation & Untraceability
  - Agent Communication Poisoning
  - Insecure Inter-Agent Protocol Abuse

Legend:
OBO: On behalf Of

Trust and Security for Agents

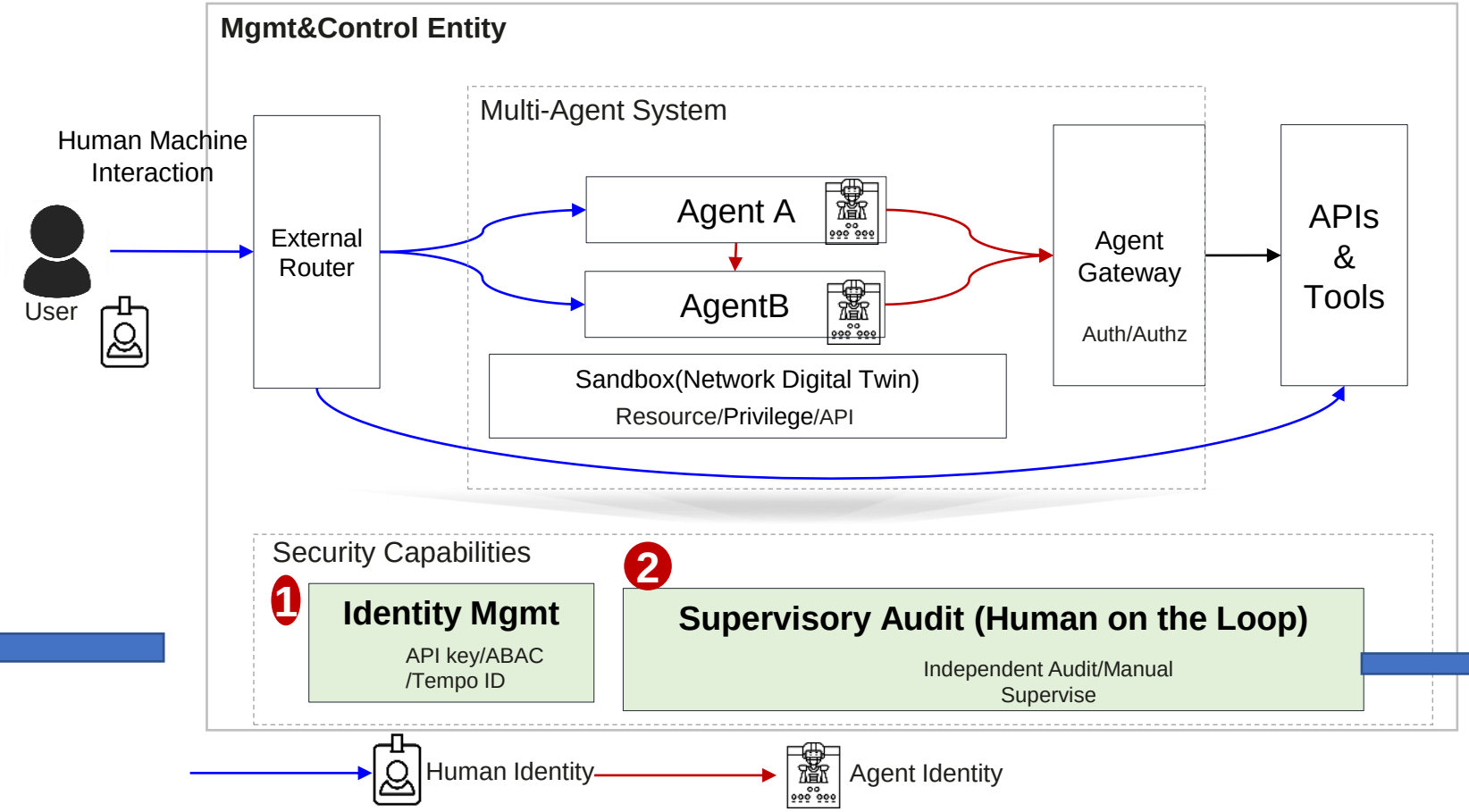
Solution to Challenge 3:

AS-IS

Static Access Control with Human ID

TO-BE

Dynamic Detection with Agent ID



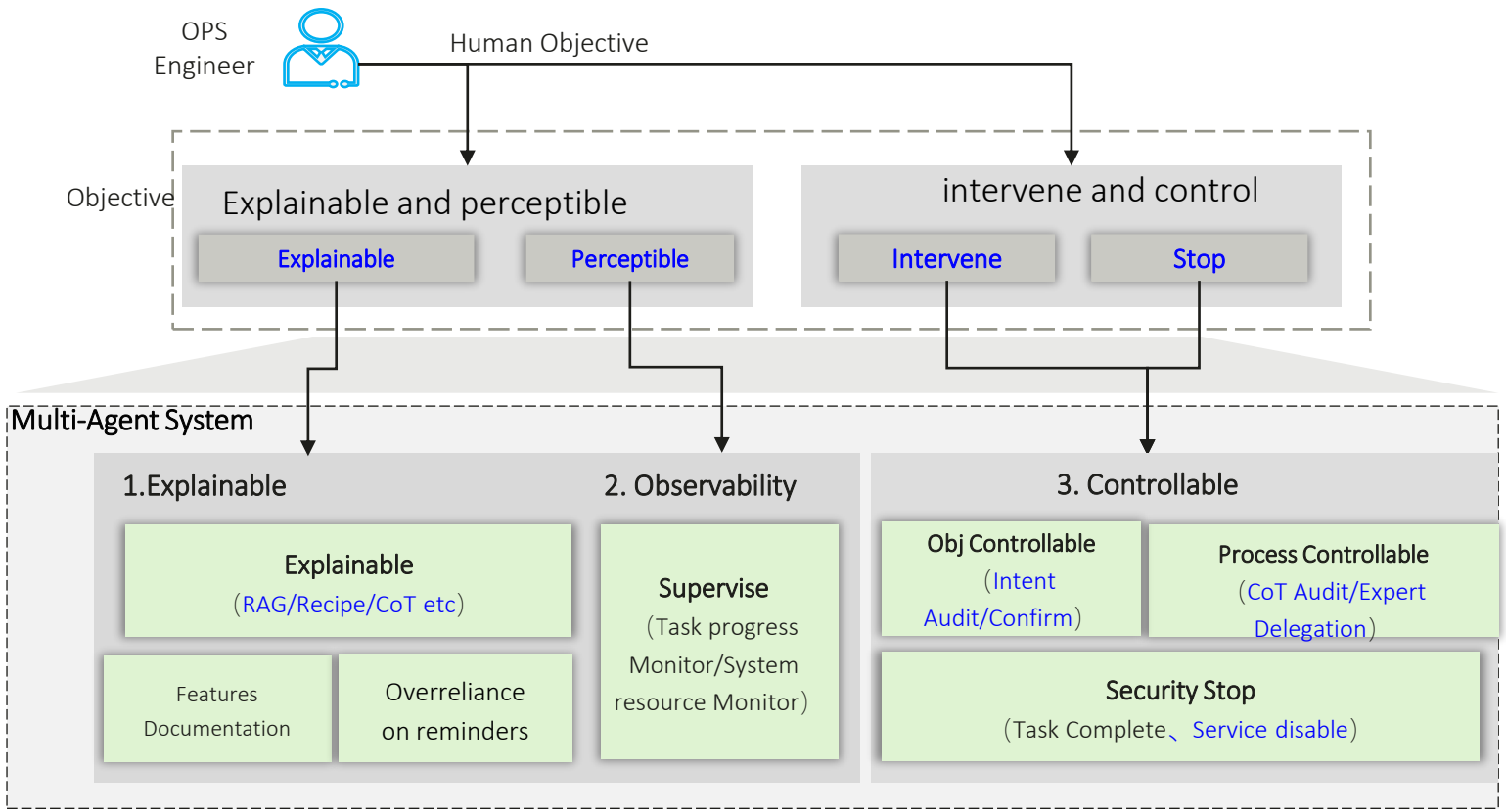
SCIM: user identity across Cloud domain
WIMSE: Workload identity
Webboauth: Bot Identity or Agent Identity
RATS: Attestation on system's identity
Oauth:Identity federation

Where should Human on the Loop be done in IETF?

Human on the Loop

Solution to Challenge 3:

Human Supervision



Key measures

Capabilities	Measures
Explainable	RAG Call Back
	Static provisioned Recipe In the prompt template
	Dynamic Planning CoT With steps and execution results
Observability	LLM Reasoning metrics Monitoring and alarm
	Agent Monitoring for trace, log, snapshot, service metrics
	System resource Monitoring such as mem, CPU, redis, etc
Controllable	Intent Audit with multi-step dialogue and human confirm
	Procedure Intervene with high risk operation audit, workflow pause/resume,
	One Click Shutdown

Candidate IETF work:

- Human and Agent Interaction to support Explainable, Observability and Controllable
  - Monitor outcome, enforce governance policies and intervene only for exceptional or high risk situation
  - Interact with Agent through natural language
  - Allow operators to review decision, request explanations and provide guidance when necessary

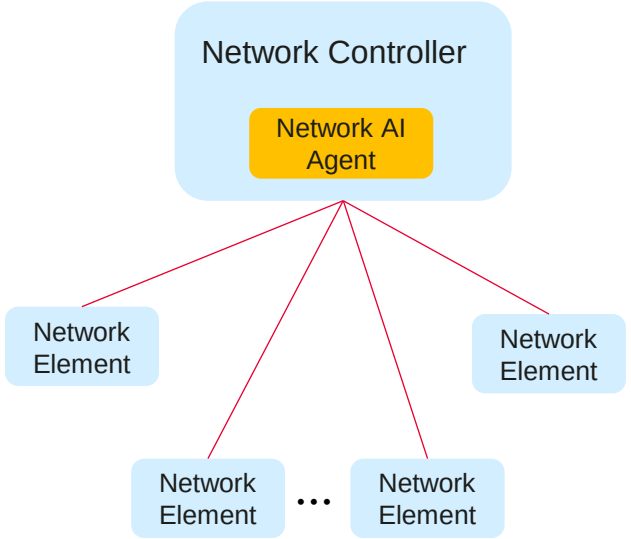
Potential Work Items in IETF

- **Starting point** for IETF agneticOPS work
  - Nature Language Interaction protocol to ensure
    - Both The accuracy and efficiency of structured data for deterministic tasks
    - and natural language interactions for understanding and handling uncertain or ambiguous tasks.
  - Translate Service Level YANG Data model and Network Level YANG Data model into DSL payloads and APIs
    - which can be consumed by Agents and Models.
  - Human and Agent Interaction to support
    - Explainable
    - Observability
    - controllable capabilities.
  - Define semantic information transfer or agent prompt language template
    - for Agent to Agent Communication on IETF intent
  - Agent to Agent Protocol extensions
    - E.g., Policy management for IP Network Agent and Network Element Agent Collaboration.

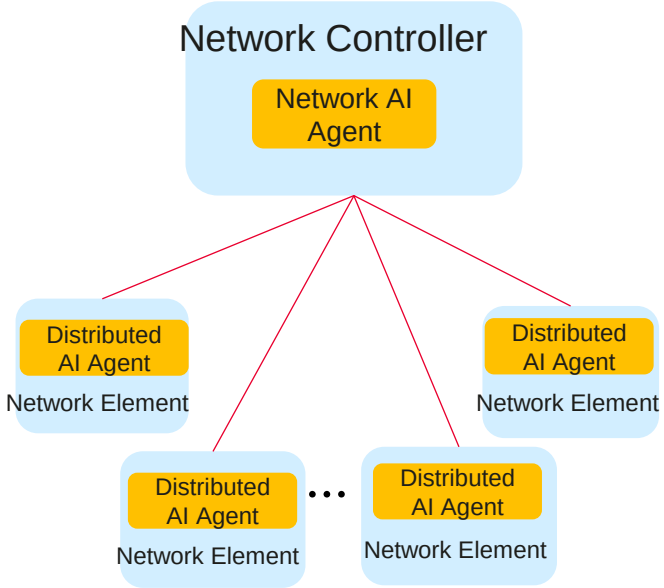
Appendix:
Agent to Agent Protocol extensions in NM

Evolution Toward Distributed Network Agent Architecture

Mode 1: Centralized Network Agent Architecture



Mode 2: Distributed Network Agent Architecture



Challenges

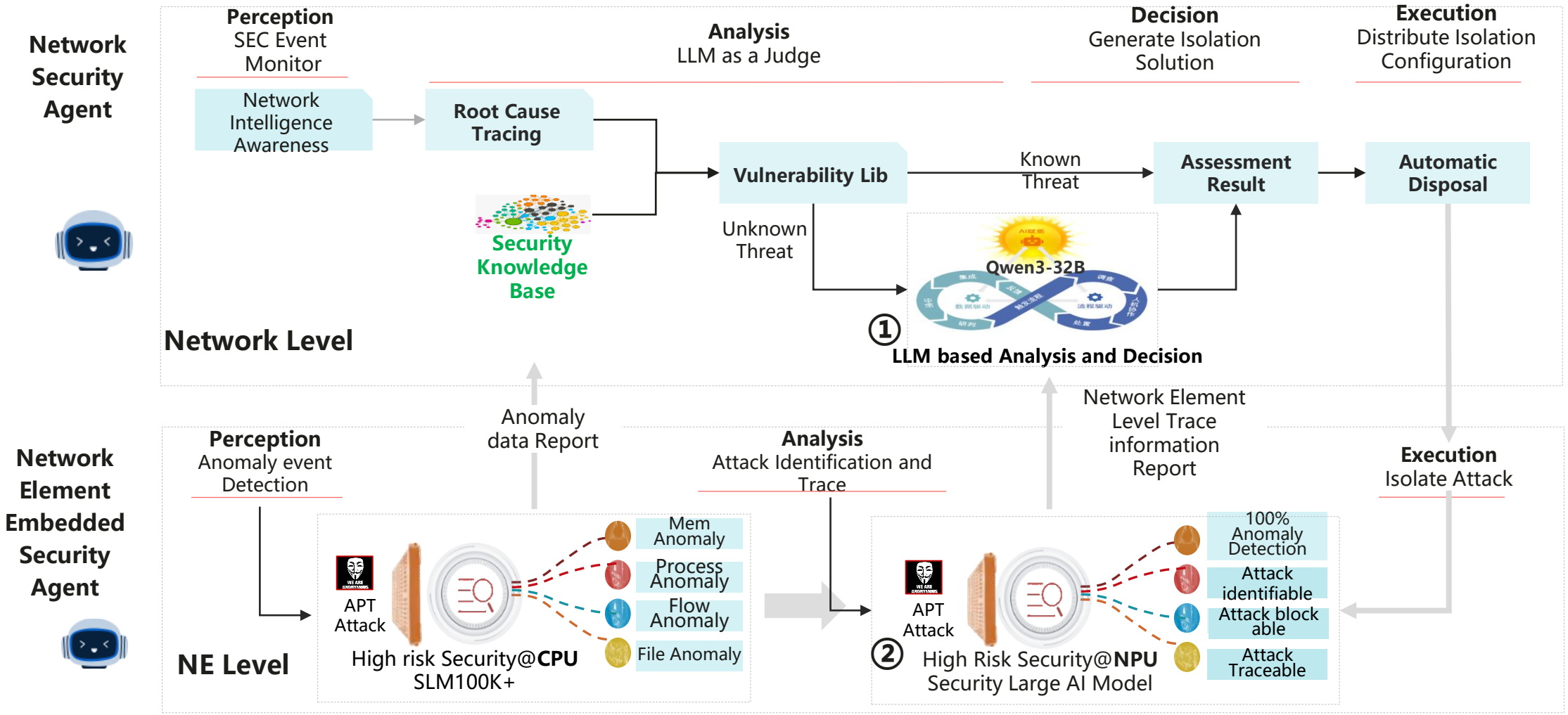
High Velocity & Latency for Telemetry

Multiple Transaction Rounds e.g.,20+ interaction with Network Element

Complex Orchestration 5 tasks with each having 10+ API invoking

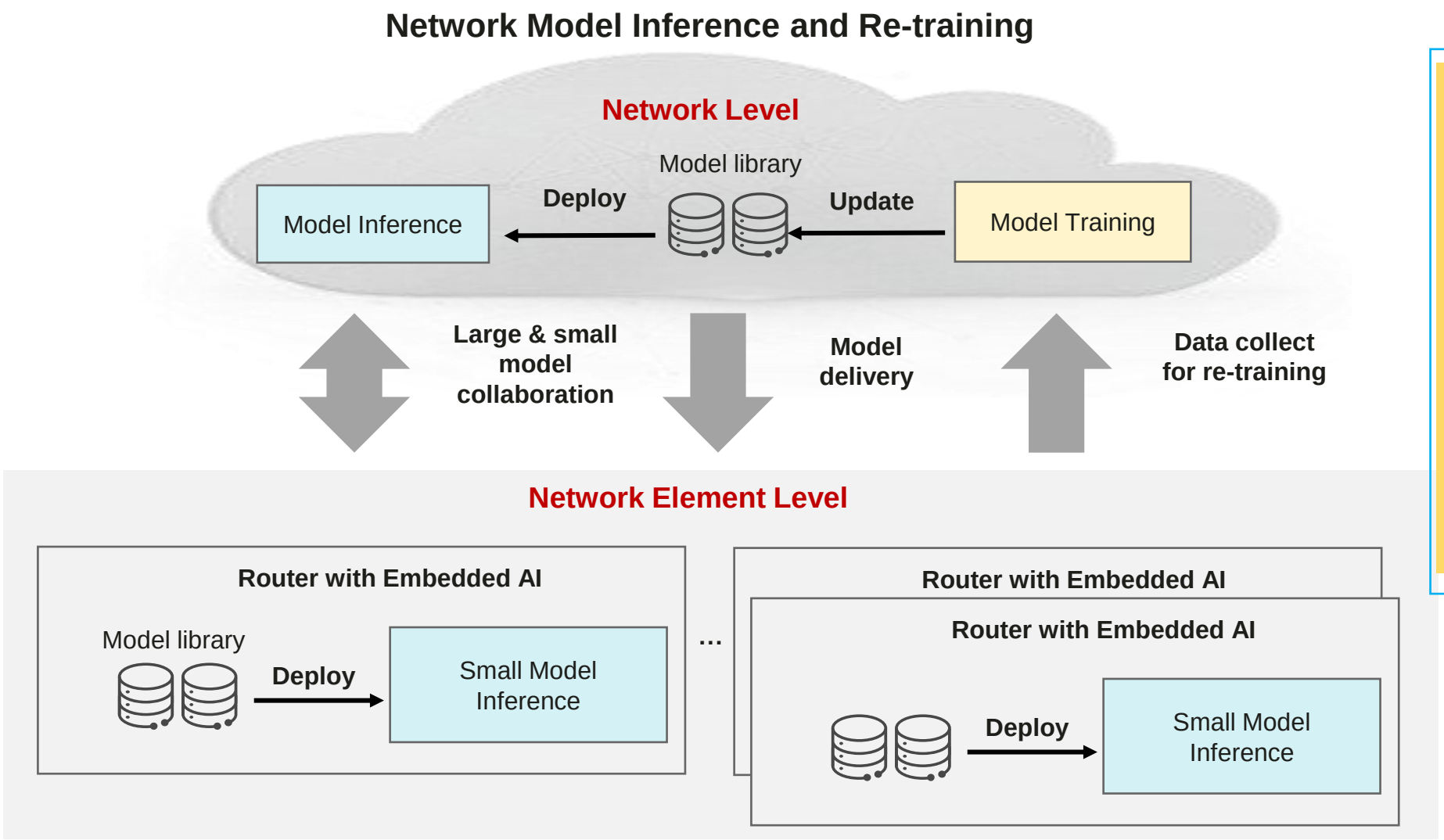
Silo data from heterogeneous Data Source with Noise

Domain Specific Autonomous With Multi-Security Agents Collaboration



Network AI Agent and NE AI Agents Collaboration

Domain Specific Autonomous IP Network Agent and Network Element Agent Collaboration



Candidate IETF work:

- Agent to Agent Protocol extensions for IP Network Agent and Network Element Agent Collaboration or policy management
- Enabling Network devices as MCP server
- allow AI models to interact with network equipment using natural language and structured tools